

Randomized Controlled Trial to Evaluate the Impact of Pharmaceutical Care on Therapeutic Success in HIV-Infected Patients in Southern Brazil

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Abstract A non-blinded randomized controlled trial evaluated efficacy of pharmaceutical care (PC) (Dáder method) on self-reported antiretroviral adherence and undetectable plasma viral load (UPVL), compared with usual care (UC) in HIV-positive patients in Brazil. Most were male (63 %), mean age 40.3 years (SD = 10). After 12 months, 79.8 % of those receiving PC versus 73.8 % in UC were adherent (RR 1.05, 95 % CI 0.95–1.15, $P = 0.35$), and 50.3 versus 49.8 % (PC vs. UC, respectively) had UPVL (RR 1.08, 95 % CI 0.97–1.20, $P = 0.15$). Factors associated with self-reported adherence were regular employment, UPVL, no depressive symptoms, and lower pill load in the treatment regimen. Older age, education, CD4 count <200 cells/mm³, and shorter treatment duration were associated with UPVL. Of 94 reported drug-related problems, 43 % resolved. In subgroup analyses of patients who experienced therapeutic failure and who were non-adherent at the beginning of the trial ($N = 50$), no differences were found in UPVL (HR

1.35, 95 % CI 0.57–3.19). Only education level (>12 years) was independently predictive of viral suppression (HR 7.47, 95 % CI 1.69–32.91). In conclusion, PC was not associated with increased self-reported adherence to ART or UPVL in patients treated at a health care facility in southern Brazil. The study suggests, however, that PC could be effective for the subgroup of patients with poor adherence.

Resumen Se evaluó, a través de ensayo clínico abierto, la eficacia de la Atención Farmacéutica (AF) (Método Dáder) sobre la adherencia auto-reportada a la terapia antirretroviral y obtención de carga viral indetectable (CVi), comparada con cuidado habitual (CH), en pacientes HIV-positivos en Brazil. La mayoría fue hombres (63 %), edad media de 40.3 (DP = 10) años. Después de 12 meses, 79.8 % del grupo AF frente 73.8 % en CH estaban adherentes (RR = 1.05, 95 % CI 0.95–1.15, $P = 0.35$), y 50.3 frente 49.8 % (AF vs. CH, respectivamente) tenían CVi (RR = 1.08, 95 % CI 0.97–1.20, $P = 0.15$). Los factores asociados con adherencia auto-reportada fueron empleo regular, CVi, sin síntomas depresivos y menor número de comprimidos. Ser mayor, educación, CD4 <200 cel/mm³ y menor duración de tratamiento, se asociaron con CVi. De los 94 problemas relacionados a los medicamentos encontrados, 43 % fueron resueltos. En la análisis de subgrupo con pacientes que habían experimentado fracaso terapéutico y no adhesión al principio del ensayo ($N = 50$), tampoco se encontró diferencia (HR 1.35, 95 % CI 0.57–3.19). Apenas educación >12 años fue predictor independiente de CVi (HR 7.47, 95 % CI 1.69–32.91). En conclusión, AF no se mostro asociada con aumento de la adherencia auto-reportada o CVi entre los pacientes de este estudio. Entretanto, los resultados sugieren que AF puede ser efectiva para un subgrupo de pacientes con poca adherencia.

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Palabras-Clave Atención farmacéutica · Ensayo clínico aleatorio · Terapia antirretroviral · VIH · Adherencia

Introduction

Pharmaceutical care (PC) is a professional pharmaceutical practice aimed directly at patients, wherein pharmacist skills, attitudes, behaviors, commitments, concerns, ethics, functions, knowledge, and responsibilities are focused on the provision of drug therapy to achieve optimal therapeutic and quality of life outcomes [1]. The components of professional practice for the exercise of PC are monitoring/pharmacotherapeutic follow-up, pharmaceutical education, dispensation, pharmaceutical services, systematic registry of activities, and measurement and evaluation of the results [2]. In this sense, the pharmacist becomes responsible for the patient's pharmacotherapeutic results, seeking early identification of drug-related problems in order to prevent and solve them.

Identification and resolution of drug-related problems (DRP), cost reduction [3], and high levels of patient satisfaction are associated with PC [4]. Ma et al. [5] demonstrated that clinical pharmacist interventions increased antiretroviral therapy (ART) adherence in 81–89 % of patients. Meta-analysis of 36 RCTs evaluating the effect of interventions on adherence and viral load suggests that the interventions were effective regardless of the kind of standard or usual care (UC) [6]. The majority of ART interventions are nurse-led, and few trials of PC in HIV patients have been conducted.

In this study, we assessed the efficacy of PC, specifically the Dáder method, on therapeutic success in HIV-infected patients in southern Brazil. We hypothesized that PC would improve adherence overall, especially among patients with early low-adherence or with detectable viral load. This project adds to current knowledge regarding how pharmacist-delivered interventions can be implemented in clinical practice.

Methods

We conducted a non-blinded randomized controlled trial comparing PC with UC in Brazilian patients receiving ART. The trial was registered at ClinicalTrials.gov (NCT00959361), and the protocol was published [7].

Study Population and Setting

Patients were eligible for the study if they were receiving care for HIV infection, including ART, at the Service for Specialized Assistance in HIV (SAEH) at the School of Medicine–UFPEL, in Pelotas, southern Brazil. ART was provided by the Brazilian public health system to all patients. Inclusion criteria for this trial included: patients aged 18 years or older, living in the urban area of Pelotas, and not pregnant. Exclusion criteria included mental illness, vision disease, and hospitalization or terminal illness, based on clinical records. We excluded those who indicated during in-person screening that they would not be staying or residing in the area during the 1-year trial duration. Exclusion criteria also included being able to participate in interviews and being able to see the color-coded chart used in PC. Patients with documented mental illness (including Alzheimer's disease and dementia) were excluded so as not to introduce confounding variables into the interviews, situational analysis, and evaluation. Consenting participants were randomized using a random number table.

Intervention

The PC intervention was administered monthly by a pharmacist utilizing the Dáder method [8], for 12 months. In brief, the PC intervention involved patients receiving structured counseling on their prescription regimens, at the time of initial drug dispensing and at monthly refill visits (Fig. 1). The key elements of the PC counseling were [1] reviewing the prescription with the patient [2]; reviewing a card on which medications were color-coded to facilitate recognition and reduce confusion that might arise from complicated drug names [3]; reviewing the schedule, length, and date of the next appointment [4]; reviewing the patient's understanding of the prescription by asking him/her to describe it for the pharmacist [5]; and giving patients verbal information on the expected side effects of their medications and instructing them to seek medical assistance by calling the pharmacist if side effects occurred. After the counseling session, the pharmacist verified that all components of the intervention had been delivered.

Brief Description of Dáder Method

The Dáder method has been described in detail previously [7, 8]. It consists of a series of scheduled meetings at which the pharmacist and patient address, review, and solve drug-related problems (DRP). The first meeting begins with open questions about health problems that concern the patient, pharmacotherapeutic history, and identification of DRP. The patient's medications and use history are reviewed, and a situational analysis and global evaluation

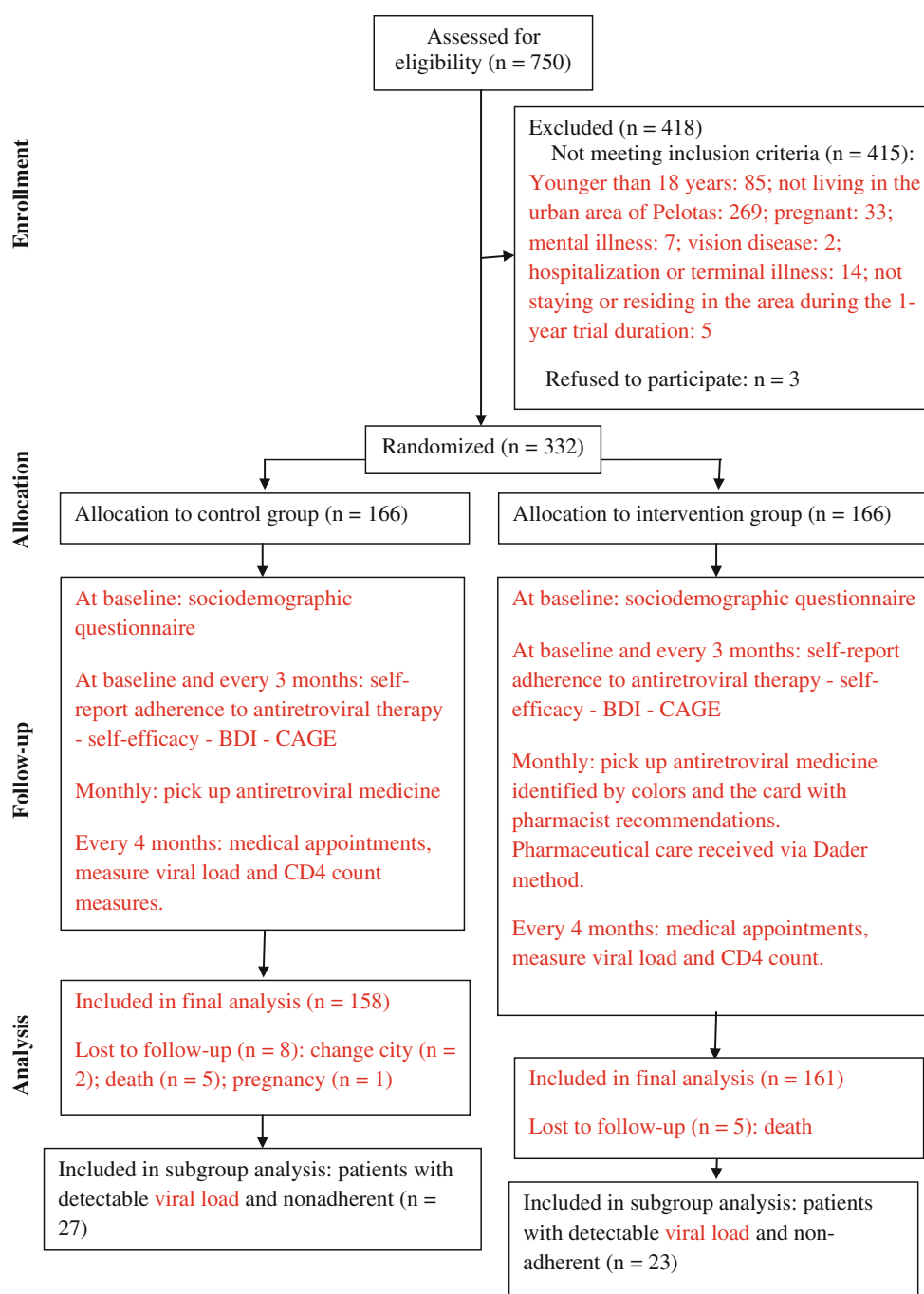


Fig. 1 Flow chart of the progress through the phases of study

are performed to identify, prevent, and solve DRP. The second meeting focuses on potential DRP, which are listed for the patient, and strategies to solve them (pharmaceutical intervention). Guidelines about drugs and adherence are provided. Values of viral load, CD4 count, and other laboratory tests (complete blood count, electrolytes, lipid profile) are reviewed. All follow-up evaluations continue to focus on solving DRP. According to the Dáder protocol, monthly meetings are not necessary unless there is a

change in treatment or the patient requests it. In this trial, however, all patients in the PC group received at least 1 pharmaceutical intervention monthly, for 12 months.

Identification of Drug-Related Problems

Using the Brazilian Consensus of Pharmaceutical Care [2], we defined DRP as “any health problem related or suspected to be related to drug therapy that interferes or may

interfere in the results of therapy or in the quality of life of the patient”. A DRP was considered to be present whenever there was an issue regarding need, effectiveness, or safety, or whenever these were suboptimal. DRPs were queried for and investigated by the pharmacist and physician, as needed, and were reviewed in monthly meetings. Operationally these included instances in which prescribed medications were not being taken, non-prescribed medications were being taken, or there was evidence of drug failure or side effects. During the PC visit, the pharmacist discussed possible causes and solutions for DRPs with the patient. More complex clinical problems were referred for discussion with the patient’s physician.

Control Group (UC)

The control group received UC for ART drug delivery. At the first appointment, when patients received their medications, a nurse first provided information on the regimen, how and when to use the medications, principal side effects, and the importance of adhering to the prescription. Afterward, patients picked up their medications at a drug dispensation counter. They had no encounters with a pharmacist. Patients were informed that they could schedule time with a nurse for any questions about their treatment or disease.

In both groups, patients were scheduled every 4 months for medical appointments and measurements of viral load and CD4 count. Participants who did not appear for their regularly scheduled appointment were contacted by telephone and asked to return. Fifteen were contacted by telephone 1 time (8 in the UC group, 7 in the PC group) and only 1 patient in the UC group did not return.

Measurements

Data Collection

In both the PC and UC groups, interviewers administered questionnaires at baseline (sociodemographic data) and every 3 months to assess adherence to antiretroviral drugs [9, 10], symptoms of depression (Beck Depression Inventory, BDI), and alcohol use or dependence (CAGE questionnaire) (Fig. 1). The interviews took place at SAEH before or after a scheduled visit with the physician and were conducted by trained students and pharmacists.

Variables

Adherence to ART was estimated by self-reporting, using a time sheet that prompted participants for routine activities (e.g., breakfast, lunch) and medicines taken, doses, and schedules in the 3 days preceding the interview. We

calculated adherence for each medication by taking the number of tablets that participants reported ingesting and dividing it by the number they should have ingested. We classified participants as adherent if they reported using 95 % or more of the tablets prescribed [11]. We investigated depression using a validated Portuguese-language version of BDI with standard scoring [12]. The BDI cutoff scores were: 0–11 not depressed, 12–19 mildly depressed, 20–35 moderately depressed, and 36–41 severely depressed [12]. A positive response to 2 or more items in the CAGE questionnaire was considered to be a positive screen for potential alcohol dependence. Treatment duration and regimens were abstracted from medical records. Clinical stage of HIV infection was classified according to the US Centers for Disease Control and Prevention [13] and modified according to the Rio de Janeiro/Caracas criteria. Viral load for all participants was measured periodically using the bDNA method at a single laboratory accredited by the National Health Council. The detection limit was 50 copies/mL.

Statistical Analysis

The null hypothesis stated that the proportion of patients reporting adherence to ART and with undetectable viral load would be the same for the PC group and the UC group.

Sample Size

The sample size was estimated using a 2-tailed test with significance level of 0.05, a 56 % proportion of adherent (defined as taking 95 % of prescribed ART) patients in the control arm [8, 9], and a power of 80 % to show a difference of at least 30 % (RR 1.3) in the outcome. Adding 20 % for losses and refusal, we estimated that a total of 166 patients would be needed in each group.

Data analysis

Data were entered in duplicate, and variables were described by measurements of central tendency and proportions (Chi square test). In view of the repeated measurements, the associations were analyzed using generalized linear models in SPSS 16.0 (SPSS Inc., Chicago, USA). Covariates were selected using a hierarchical model, according to our theoretical model based on epidemiological and clinical relevance of variables. At the first level, we entered socioeconomic variables (sex, age, family income, education); at second level, we added social variables (living alone, alcohol use, regular employment); at third level, clinical variables (immune status, viral load, depressive symptoms); at fourth level, treatment-related

variables (treatment period, group of antiretroviral medications, total of tablets); and, at fifth level, the intervention variable (PC), dichotomized as the intervention group or control group. Variables with $P < 0.2$ were included in the next level. Using the kappa statistic, we also examined the association between self-reported adherence and plasma viral load.

In a subgroup analysis, to further assess the impact of PC on adherence and UPVL, we excluded patients who had self-reported being adherent and experiencing therapeutic effects at recruitment. In other words, we excluded those who had experienced the desired outcomes at the beginning of the study, resulting in a subset of patients who had experienced therapeutic failure and non-adherence at the beginning of the trial (see flow diagram in Fig. 1). Prior therapeutic failure was defined as viral load >50 copies/mL, and non-adherence was defined as reporting <95 % intake of pills after at least 1 month on ART. The outcome “time to success in treatment” was defined as the number of days to the first viral load test result of fewer than 50 copies/mL. We then calculated the survival rate (point estimates and 95 % CI) for 6-month intervals using the Kaplan–Meier method. The Cox proportional hazards model was used to calculate the crude and adjusted hazard ratios for the main predictor (intervention vs. UC group). The assumption of proportionality in hazards was assessed both by log-rank survival plot and by Schoenfeld residuals test; both indicated that hazard proportionality was not violated. Bivariate interaction terms were assessed by likelihood ratio, and none were significant at $\alpha = 0.1$. The final fitted model was used to assess confounding. The criteria for virological success was a viral load of fewer than 50 copies/mL (classified as undetectable viral load). The final model in a subgroup analysis included sex, age, education, social services, and intervention group with no interaction term. For the subgroup analysis, we used STATA 12 (StataCorp, College Station, USA).

Results

Of 335 patients invited to participate, 332 (99 %) consented. Thirteen patients were lost to follow-up (Fig. 1). At baseline the majority of participants were male (63 %), with a mean age of 40.3 years ($SD = 10$), a median family income of 1.4 Brazilian minimum-wage units, and a mean of 6.6 years ($SD = 3.75$) of formal education. Sixteen participants (5 %) could not read, all had housing, and slightly more than half (55 %) lived with another person. Only 4 participants (1 %) reported having injected drugs in the last 3 months, and most (65 %) reported no use of alcohol in the last 4 weeks. Eighty-five participants (34 %) had a BDI score in the range of moderately to severely depressed, and only 1 had a positive CAGE result. At the first interview, 94 % of participants were in clinical stage C 171 (52 %) had an undetectable viral load, and 252 (76 %) had CD4 counts ≥ 200 cells/mm³. Seventy-six percent were classified as adherent. Eighty-one percent used 3 ART drugs - of these participants, 73 % were using non-nucleoside reverse transcriptase inhibitor-based combinations and 29 % were using protease inhibitor-based combinations. Overall, 175 (53 %) had switched their regimen at least once. Duration of ART ranged from 1 to 174 months, with a mean of 51.7 months ($SD = 34.8$).

Table 1 describes the baseline characteristics of patients in each group, which were mostly similar. The only statistically significant difference was that those in the intervention group were more likely to be receiving social-service benefits (Chi square test, $P = 0.004$). The proportion of patients self-reporting adherence to ART at baseline was higher than expected (79.5 % for the PC group, 72.3 % for the UC group), and after 12 months of follow-up the proportions changed to 79.8 and 73.8 %, respectively ($P = 0.95$). After 12 months of follow-up, viral load was undetectable in 50.3 % of the PC group and 49.8 % of the UC group. Tables 2 and 3 show that no significant differences persisted after controlling for potential confounding variables (self-reported adherence: RR 1.05, 95 % CI 0.95–1.15, $P = 0.35$; undetectable viral load: RR 1.08, 95 % CI 0.97–1.20, $P = 0.15$).

We examined factors associated with self-reported adherence in the overall study population using the generalized linear model. Of significance were no regular employment (RR 0.91, 95 % CI 0.82–1.00, $P = 0.05$), detectable viral load (RR 0.83, 95 % CI 0.73–0.95, $P = 0.01$), depressive symptoms (BDI ≥ 12) (RR 0.99, 95 % CI 0.99–1.00, $P = 0.04$), and high number of tablets (RR 0.96, 95 % CI 0.93–0.98, $P < 0.01$) (Table 2). Factors associated with undetectable viral load were older age (RR 1.01, 95 % CI 1.00–1.01, $P = 0.02$), years of schooling (RR 1.02, 95 % CI 1.00–1.03, $P = 0.06$), CD4 count <200 cells/mm³ (RR 0.62, 95 % CI 0.50–0.77, $P < 0.001$), and shorter duration of treatment (RR 1.00, 95 % CI 0.99–1.00, $P = 0.03$) (Table 3).

We detected 94 drug-related problems in the intervention group; 40 (43 %) of these were resolved by PC. Fifty DRPs (53 %) were related to medicine use, 20 participants did not use prescribed medicine, and 30 used no prescribed medicines. Forty DRPs were related to no effectiveness and 4 to adverse events. Forty-two patients (30 %) were referred for medical consultation. All 10 patients who were prescribed didanosine, which should be dissolved or broken, used it incorrectly. We also identified 4 dispensing errors.

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Table 1 Baseline characteristics of participants in the randomized controlled trial to evaluate the impact of pharmaceutical care on therapeutic success in HIV-infected patients, Pelotas, Brazil, 2006–2009. (in each group, $N = 166$)

	Intervention	Control	P value ^a
Sociodemographic characteristics			
Male, N (%)	108 (65.1)	102 (61.4)	0.495
Age, mean (SD)	39.28 (9.4)	41.33 (10.7)	0.065
Years of formal education, mean (SD)	6.63 (3.8)	6.67 (3.7)	0.907
Family income ^b (multiples of Brazilian minimum wage)			
Median (IQR 25–75)	1.29 (1.0–2.1)	1.70 (1.0–2.7)	0.155
No regular employment, N (%)	117 (70.5)	111 (66.9)	0.478
Received social-service benefits, N (%)	101 (60.8)	75 (45.2)	0.004
Participant housing, N (%)			0.340
Alone	28 (17.0)	30 (18.1)	
With relatives, partner	135 (81.8)	130 (78.3)	
With friends	2 (1.2)	6 (3.6)	
Alcohol consumption over the last month, N (%)			0.438
Every day	2 (1.2)	3 (1.8)	
At least once a week	28 (17.1)	24 (14.5)	
Less than once a week	33 (20.1)	24 (14.5)	
Never	101 (61.6)	114 (69.1)	
Clinical characteristics			
Antiretroviral treatment regimen, N (%)			0.714
2NRTIs + NNRTI	118 (71.1)	118 (71.5)	
2NRTIs + PI	44 (26.5)	45 (27.3)	
NRTI + NNRTI + PI or FI	4 (2.4)	2 (1.2)	
Antiretroviral regimen changes ^c , N (%)			0.772
First	77 (49.0)	80 (51.0)	
Second	39 (50.0)	39 (50.0)	
Third	24 (52.2)	22 (47.8)	
Fourth	14 (60.9)	9 (39.1)	
Fifth or more	12 (42.9)	16 (57.1)	
Undetectable viral load, N (%)	86 (52.1)	85 (51.2)	0.868
CD4 count ≥ 200 cells/mm ³ , N (%)	129 (77.7)	123 (74.1)	0.441
Depressive symptoms, N (%)			0.890
BDI < 12	121 (72.0)	117 (71.3)	
BDI ≥ 12	47 (28.0)	47 (28.7)	
Adherent to antiretroviral therapy ^d , N (%)	132 (79.5)	120 (72.3)	0.124
Time of treatment, mean (SD)	52.78 (35.9)	50.61 (33.7)	0.565
Number of tablets prescribed daily			0.096
Mean (SD)	3.31 (0.7)	3.20 (0.5)	

NRTI nucleoside reverse transcriptase inhibitor, NNRTI non-nucleoside reverse transcriptase inhibitor, PI protease inhibitor, FI fusion inhibitor

^a Chi square test

^b Minimum wage = R\$ 350–US\$ 153.51–US\$ 1 = R\$ 2.28

^c Participants can fall into more than one category

^d Patient was considered to be adherent if he/she reported use of ≥ 95 % of the prescribed dose during the 3 days of evaluation

Subgroup Analysis Results

After randomization, 50 patients who had detectable viral load and were non-adherent to ART at baseline were chosen for subgroup analysis (18.1 % of UC group and 13.9 % of PC group) (see Fig. 1). Baseline characteristics were similar in both the PC and UC groups.

After 2 years of follow-up, the cumulative proportion of participants with virological success was 76 % (95 % CI, 50.0–94.7) in the PC group and 42.9 % (95 % CI, 23.9–68.3) in the UC group (log-rank test for equality of

survivor functions: $\chi^2 = 1.24$, $P = 0.266$) (Fig. 2). The Kaplan–Meier success-estimate curves were not statistically significant (log-rank test 1.24, $P = 0.266$) (Fig. 3).

In bivariate analysis, 12 or more years of formal education was associated with virological success and persisted (HR 7.47, 95 % CI, 1.69–32.9, $P = 0.008$) after adjusting for sex, social benefit and group of treatment. Patients in the PC group had an insignificant 35 % higher chance of virological success than patients in the UC group (HR 1.35, 95 % CI, 0.57–3.19, $P = 0.499$).

Table 2 Variables associated with self-reported adherence of ≥ 95 % at follow-up (generalized linear models) among participants in a randomized controlled trial to evaluate the impact of pharmaceutical care on therapeutic success in HIV-infected patients, Pelotas, Brazil, 2006–2009

Level	Variables	Crude relative risk (95 % CI) ^a	<i>P</i> value	Adjusted relative risk (95 % CI) ^{a,b}	<i>P</i> value
1	Sex, male	0.97 (0.93–1.02)	0.25	0.96 (0.88–1.06)	0.44
	Age ^c	1.00 (0.99–1.01)	0.97	1.00 (0.99–1.00)	0.81
	Family income ^c	1.00 (0.99–1.02)	0.74	1.00 (0.97–1.02)	0.83
	Years of formal education ^c	1.01 (1.00–1.01)	0.07	1.01 (1.00–1.02)	0.17
2	Shared residence				
	Yes	1.01 (0.95–1.08)	0.72	0.95 (0.83–1.07)	0.40
	Frequency of alcohol consumption		0.19		0.74
	Every day	0.93 (0.77–1.12)	0.43	0.94 (0.59–1.49)	0.79
	At least once a week	1.04 (0.97–1.12)	0.23	1.02 (0.91–1.15)	0.74
	Less than once a week	1.06 (0.99–1.13)	0.08	1.06 (0.95–1.18)	0.28
	Never	1		1	
	No regular employment	0.93 (0.88–0.98)	<0.01	0.91 (0.82–1.00)	0.05
3	CD4 count <200 cells/mm ^{3d}	0.94 (0.88–1.01)	0.08	0.98 (0.85–1.13)	0.81
	Detectable viral load ^d	0.89 (0.84–0.94)	<0.01	0.83 (0.73–0.95)	0.01
	Depressive symptoms (BDI ≥ 12) ^d	1.00 (0.99–1.00)	<0.01	0.99 (0.99–1.00)	0.04
4	Treatment duration	1.00 (0.99–1.00)	<0.01	0.99 (0.99–1.00)	0.39
	ART ^e regimen				0.37
	2NRTIs ^f + NNRTI ^g	1.22 (1.02–1.47)	0.03	0.84 (0.55–1.28)	0.42
	2NRTIs + PI ^h	1.09 (0.90–1.31)	0.37	0.93 (0.63–1.39)	0.73
	NRTI + NNRTI + PI or FI ⁱ	1		1	
	Number of tablets daily	0.98 (0.97–0.98)	<0.01	0.96 (0.93–0.98)	<0.01
5	Pharmaceutical care				
	Intervention group	1.03 (0.99–1.09)	0.17	1.05 (0.95–1.15)	0.35

BDI beck depression inventory

^a Relative risk and confidence interval of 95 % calculated by regression for generalized linear models^b Final model: years of formal education, regular employment, viral load, depressive symptoms, and total number of tablets^c Continuous variables^d Compared with CD4 count ≥ 200 cells/mm³, undetectable viral load, and BDI <12, respectively^e Antiretroviral therapy^f Nucleoside reverse transcriptase inhibitor^g Non-nucleoside reverse transcriptase inhibitor^h Protease inhibitorⁱ Fusion inhibitor

Discussion

In this randomized controlled trial to assess the effectiveness of PC on adherence to ART, it was found that an enhanced intervention did not produce any additional benefit over current standard practices. The null results of the study, plus the overall high adherence in the study population, suggest that current procedures providing basic nursing and medical care support for adherence are adequate. Similarly, Echevarría et al. [14] conducted a multi-center cohort study and found no statistically significant difference in ART adherence with PC. In a review and synthesis of adherence studies [15], it was noted that

studies targeting those with known or anticipated adherence problems showed significantly greater intervention effects than studies that did not target poor adherers. The authors stressed the importance of carefully planning these studies and their inclusion criteria because patients with adherence problems would benefit more from intervention programs. The more intensive PC that we tested may need to be considered for those who benefit less from UC—for example, patients who experience more side effects or who have problems with substance use or alcohol, which are known to affect adherence.

Overall baseline adherence in our sample was high (76 %), which may have contributed to the insignificant

Table 3 Variables associated with undetectable viral load at follow-up (generalized linear models) among participants in a randomized controlled trial to evaluate the impact of pharmaceutical care on therapeutic success in HIV-infected patients, Pelotas, Brazil, 2006–2009

Level	Variables	Crude relative risk (95 % CI) ^a	P value	Adjusted relative risk (95 % CI) ^{a,b}	P value
1	Sex, male	0.98 (0.93–1.04)	0.48	0.96 (0.85–1.09)	0.52
	Age ^c	1.00 (1.00–1.01)	0.04	1.01 (1.00–1.01)	0.02
	Family income ^c	0.99 (0.98–1.01)	0.37	0.98 (0.94–1.02)	0.40
	Formal education ^c	1.01 (1.00–1.02)	0.03	1.02 (1.00–1.03)	0.06
2	Shared residence				
	Yes	1.01 (0.93–1.08)	0.88	0.91 (0.77–1.08)	0.29
	Frequency of alcohol consumption		0.29		0.69
	Every day	1.01 (0.82–1.24)	0.92	1.13 (0.71–1.78)	0.61
	At least once a week	0.93 (0.86–1.00)	0.06	0.99 (0.75–1.09)	0.29
	Less than once a week	0.99 (0.92–1.07)	0.75	0.99 (0.84–1.17)	0.93
	Never	1		1	
	No regular employment	0.97 (0.91–1.03)	0.30	0.95 (0.83–1.09)	0.45
3	CD4 count <200 cells/mm ^{3d}	0.68 (0.63–0.73)	<0.001	0.62 (0.50–0.77)	<0.001
	Depressive symptoms (BDI ≥12) ^d	1.00 (0.99–1.00)	0.03	1.00 (0.99–1.00)	0.63
4	Treatment duration	1.00 (0.99–1.00)	0.02	1.00 (0.99–1.00)	0.03
	ART ^e regimen		<0.001		0.40
	2NRTIs ^f + NNRTI ^g	1.43 (1.19–1.71)	<0.001	1.04 (0.71–1.52)	0.86
	2NRTIs + PI ^h	1.19 (0.99–1.43)	0.07	0.93 (0.62–1.40)	0.73
	NRTI + NNRTI + PI or FI ⁱ	1		1	
	Number of tablets daily ^c	0.97 (0.97–0.98)	<0.001	0.99 (0.97–1.01)	0.17
5	Pharmaceutical care				
	Intervention group ^j	1.06 (1.00–1.12)	0.05	1.08 (0.97–1.20)	0.15

^a Relative risk and confidence interval of 95 % calculated by regression for generalized linear models^b Final model: age, formal education, immune status, treatment duration, and total number of tablets^c Continuous variables^d Compared with CD4 count ≥200 cells/mm³ and BDI <12, respectively^e Antiretroviral therapy^f Nucleoside reverse transcriptase inhibitor^g Non-nucleoside reverse transcriptase inhibitor^h Protease inhibitorⁱ Fusion inhibitor^j Compared with usual care

results. Demographic and clinical characteristics in our study were similar to other studies [14, 16, 17]; however, convenience sampling at the clinic may have contributed to this: patients who regularly attend clinical services may be more adherent and responsive to treatment and, as a result, less likely to benefit from PC. We based our sample size on detecting a difference of 30 % from a baseline of 56 % adherence [9, 10].

A subgroup analysis showed that among non-adherent patients with detectable viral load, more patients in the PC group achieved undetectable viral load, suggesting that PC can be effective in higher-risk patients. Future research on the effects of PC might be better focused on these higher-risk patient populations.

A clinical trial that assessed self-reported adherence to ART over the prior 7 days [18] found a significant decrease of adherence over the 12-month study periods, falling from 74 to 67 %. About 50 % of participants were lost to follow-up (1095–540), unlike in our study, where adherence remained stable for 1 year and only a small number were lost to follow-up. Factors associated with adherence and undetectable viral load in our study are in agreement with the literature - i.e., regular employment, no symptoms of depression [19–21], fewer tablets, older age [22], more years of formal education, better immune status [22–25], and shorter treatment duration [10].

The most frequent and easily solved drug-related problems were due to patients' needs: if patients were using medications that were unnecessary, they were asked to

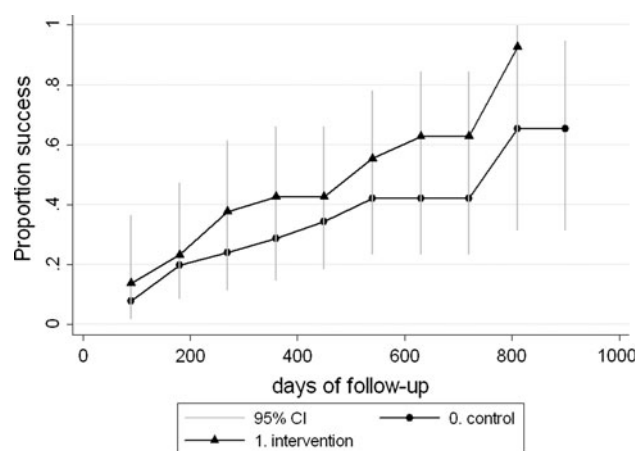


Fig. 2 Proportion of success in each group

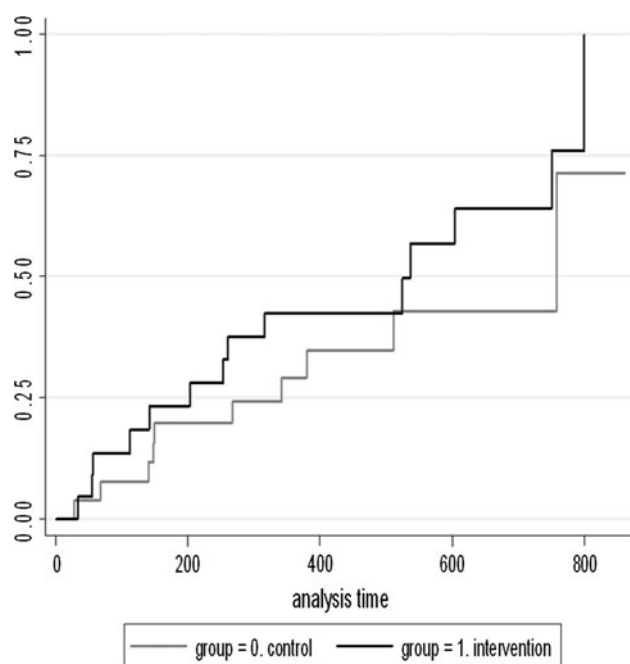


Fig. 3 Kaplan–Meier success estimates

discontinue use immediately, or, if they lacked a needed drug, a physician was asked to provide it. With regard to DRPs resulting from ineffective responses, patients receiving rescue therapy had little chance of achieving undetectable viral load, so solutions were not possible. DRPs due to safety concerns can be solved with PC, or, when this is not possible, PC can be used to motivate patients to adhere, even in the presence of adverse reaction, if continuation of the medication is clinically necessary. Several DRPs could be resolved by PC, such as counseling to increase adherence to pharmacologic and non-pharmacologic treatments and providing education about health and the correct use of medications [25].

To the best of our knowledge, this is the first randomized clinical trial using the Dáder method for HIV patients,

and some limitations should be cited. Our sample size, in retrospect, may have been inadequate to detect small differences between the PC and UC groups. Additionally, although interviewers were trained specifically to not deliver any component of PC during the interviews, we cannot exclude the possibility that some interviewers contaminated the UC group. To circumvent this bias, however, the evaluator who measured adherence did not know the ART regimen. Because viral load is an objective measurement, it is unlikely to influence interviewers. Self-reported adherence has potential to introduce bias due to overestimation, yet this measurement remains widely used and we have no reason to expect that there would be differential reporting by treatment group [9, 10, 26].

Because many authors consider viral load to be an indirect method of measuring adherence to antiretroviral medications, we analyzed the correlation between self-reported adherence and viral load. There was no concordance between the measurements. This may have been due to the fact that though patients were adherent to treatment, if they were under rescue scheme or had resistance to antiretroviral drugs, they may have had detectable viral load. On the other hand, patients may have had undetectable viral load and may not have been adherent in the days when adherence was measured.

Conclusion

We found that PC was not associated with increased self-reported adherence to ART or with an increased proportion of patients having undetectable viral load after a 12-month intervention. Additional measurements focusing on drug-related problems do not appear to produce additional benefits over basic support that includes monthly medical visits and antiretroviral drugs provided to patients by the public health system. Studies of groups at high risk of non-adherence and virological failure are needed to evaluate whether PC can increase virological success in HIV-positive adult patients undergoing ART.

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